

Week 5: Potential Energy and Energy Conservation – Tutorial Sheet

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Use the Master Procedure: identify, sketch the states and choose the reference level, sort the forces into U and W_{other} , predict, select the method, execute $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$, evaluate. Use $g = 9.80 \text{ m/s}^2$. Problems A1 and A2 are solved in full; final answers to the others are at the end.

A. Worked examples

A1. A skier on a smooth slope

A 60.0 kg skier starts from rest at the top of a smooth ($\mu = 0$) slope, 40.0 m above the valley floor. Find her speed at the bottom, and at the point 15.0 m above the valley floor. Does her mass matter?

Solution. Take $y = 0$ at the valley floor. Only gravity does work (the normal force is perpendicular to the motion), so $K_1 + U_1 = K_2 + U_2$ with $K_1 = 0$: $mg y_1 = \frac{1}{2}mv^2 + mg y_2$, hence $v = \sqrt{2g(y_1 - y_2)}$. Bottom: $v = \sqrt{2(9.80)(40.0)} = 28.0 \text{ m/s}$. At 15.0 m: $v = \sqrt{2(9.80)(25.0)} = 22.1 \text{ m/s}$. The mass cancels, so it does not matter.

A2. A block, friction and a spring

A 2.00 kg block slides at 3.00 m/s along a rough floor ($\mu_k = 0.20$). After 1.50 m it meets a light horizontal spring ($k = 400 \text{ N/m}$, the floor under the spring is also rough). Find the maximum compression of the spring.

Solution. Friction force $f_k = \mu_k mg = 0.20(2.00)(9.80) = 3.92 \text{ N}$. Take state 1 at the start ($K_1 = \frac{1}{2}(2.00)(3.00)^2 = 9.00 \text{ J}$) and state 2 at maximum compression x ($K_2 = 0$, $U_{\text{el}} = \frac{1}{2}kx^2$). The total sliding distance is $1.50 + x$: $K_1 + W_f = U_{\text{el},2} \Rightarrow 9.00 - 3.92(1.50 + x) = 200x^2$, i.e. $200x^2 + 3.92x - 3.12 = 0$. $x = \frac{-3.92 + \sqrt{3.92^2 + 4(200)(3.12)}}{2(200)} = 0.115 \text{ m}$. Check: spring energy $\frac{1}{2}(400)(0.115)^2 = 2.67 \text{ J}$; friction removed $3.92(1.615) = 6.33 \text{ J}$; total 9.00 J. ✓

B. Practice

B1. A stone in a vertical circle

A 0.300 kg stone on a string of length 0.800 m moves in a vertical circle. Its speed at the lowest point is 7.00 m/s. Find (a) its speed and (b) the tension at the top of the circle. (c) What is the smallest speed at the bottom for which the string stays taut all the way round?

B2. A diver

A 65.0 kg diver steps off a 10.0 m platform. (a) Find her speed on reaching the water. (b) She stops 3.00 m below the surface. Find the average force the water exerts on her.

B3. A hanging spring

A 0.500 kg mass hangs at rest from a light vertical spring, which stretches 0.100 m. The mass is pulled down a further 0.150 m and released from rest. Find the spring constant and the speed of the mass as it passes the equilibrium position.

B4. Down a rough incline onto a spring

A 3.00 kg block is released from rest on a 35° incline ($\mu_k = 0.30$) and slides 2.00 m before it meets a light spring ($k = 800 \text{ N/m}$) lying along the incline. Find the maximum compression of the spring.

B5. Force and energy diagram

A 0.400 kg particle has potential energy $U(x) = \frac{4.0}{x^2} - \frac{8.0}{x}$ (joules, $x > 0$ in metres). (a) Find the force at $x = 0.50 \text{ m}$ and at $x = 2.0 \text{ m}$. (b) Find the equilibrium position and state whether it is stable. (c) For $E = -2.0 \text{ J}$, find the turning points and the greatest speed.

B6. Challenge: the method is not given

A small block is released from rest at a height of 2.00 m on a smooth track. At the bottom the track becomes a horizontal rough strip of length d ($\mu_k = 0.300$), followed by a smooth vertical loop of radius 0.500 m. What is the greatest d for which the block still goes all the way round the loop?

C. Concept questions

1. Why can the gravitational potential energy of a ball be negative? Is the work done by gravity changed by the choice of $y = 0$?
2. Two frictionless ramps of different slope reach the same height. A block released from rest at the top of each reaches the bottom with the same speed. Why is this not true if friction acts?
3. A person walks up a staircase at constant speed. Which energy transformations occur? Which force does the positive work on the person?
4. On an energy diagram the total energy line lies below a maximum of U . What does that say about the particle's motion?

Exam-style practice

Ten five-option items in the style of the course examination, **suggested time: 3 minutes per item**. For each item sketch first, predict the sign or size, then calculate (IDENTIFY, SET UP, EXECUTE, EVALUATE). Choose one answer, A to E. Items marked as sharing information are linked. Answers and brief solutions are at the end of this sheet. Take $g = 9.80 \text{ m/s}^2$.

E1. A skier starts from rest at the top of a smooth (frictionless) slope and descends a vertical height of 4.5 m. Her speed at the bottom is

- (A) 4.7 m/s
- (B) 6.6 m/s
- (C) 9.4 m/s
- (D) 44.1 m/s
- (E) 88.2 m/s

Items E2 and E3 share this information: A 3.0 kg block is released from rest at the top of a 2.5 m long incline of angle 30° . The coefficient of kinetic friction is 0.25.

E2. The speed of the block at the bottom of the incline is

- (A) 3.73 m/s
- (B) 4.95 m/s
- (C) 5.93 m/s
- (D) 6.06 m/s
- (E) 7.00 m/s

E3. The thermal energy produced by friction during the descent is

- (A) 36.7 J (B) 20.8 J (C) 18.4 J (D) 15.9 J (E) 9.2 J

E4. A spring of force constant 600 N/m is compressed by 0.10 m and launches a 0.15 kg ball along a smooth horizontal surface. The speed of the ball as it leaves the spring is

- (A) 6.32 m/s
- (B) 8.94 m/s
- (C) 12.65 m/s
- (D) 40.00 m/s
- (E) 60.00 m/s

E5. Which of the following statements is/are true?

- I. The work done by a conservative force around a closed path is zero.
- II. At a minimum of the potential-energy curve $U(x)$ the net force is zero and the equilibrium is stable.
- III. The force is related to the potential energy by $F_x = -dU/dx$.

- (A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II and III

E6. A particle has potential energy $U(x) = 3.0x^2 - 2.0x^3$ (U in J, x in m). The force F_x on it at $x = 0.50$ m is

- (A) 3.0 N (B) 1.5 N (C) 0.5 N (D) -1.5 N (E) -3.0 N

E7. A roller-coaster car is released from rest on a frictionless track and then enters a vertical loop of radius 8.0 m. The least release height above the bottom of the loop for which the car stays on the track at the top is

- (A) 8.0 m (B) 12.0 m (C) 16.0 m (D) 20.0 m (E) 24.0 m

Items E8 and E9 share this information: A 0.50 kg pendulum bob on a light string of length 1.20 m is released from rest with the string at 45° from the vertical.

E8. The speed of the bob at the lowest point is

- (A) 1.86 m/s
(B) 2.62 m/s
(C) 4.08 m/s
(D) 4.85 m/s
(E) 5.25 m/s

E9. The tension in the string as the bob passes the lowest point is

- (A) 9.80 N (B) 7.77 N (C) 4.90 N (D) 2.87 N (E) 2.03 N

E10. A 2.0 kg object is dropped from a height of 10.0 m and strikes the ground at 12.0 m/s. The mechanical energy dissipated by air resistance is

- (A) 26.0 J (B) 52.0 J (C) 144 J (D) 196 J (E) 340 J

Answers

- B1 (a) 4.20 m/s; (b) 3.7 N; (c) $v_{\min} = \sqrt{5gL} = 6.26$ m/s
 B2 (a) 14.0 m/s; (b) 2.76×10^3 N (about 4.3 times her weight)
 B3 $k = 49.0$ N/m; $v = 1.48$ m/s (equals $\sqrt{k/m} \times 0.150$)
 B4 $x = 0.232$ m (quadratic: $400x^2 - 9.64x - 19.3 = 0$)
 B5 (a) $F = +32$ N at 0.50 m (toward $+x$), $F = -1.0$ N at 2.0 m; (b) $x = 1.0$ m, $U = -4.0$ J, stable; (c) $x = 2 \pm \sqrt{2} = 0.59$ m and 3.41 m; $v_{\max} = 3.16$ m/s at $x = 1.0$ m
 B6 $d = 2.5$ m (needs $v_{\text{top}}^2 \geq gR$; speed at the bottom of the loop 4.95 m/s)
 C1 reference level is a choice; $W = -\Delta U$ is unchanged
 C2 friction depends on the path length; the steeper ramp is shorter
 C3 chemical $\rightarrow U_{\text{grav}}$ and thermal; the force of the stairs does no work on the person; the muscles supply the energy
 C4 $K = E - U < 0$ at the maximum, so the particle cannot pass: it is confined to one side

Exam-style practice: answer key and brief solutions

Key: E1 C E2 A E3 D E4 A E5 E E6 D E7 D E8 B E9 B E10 B

E1 (C). Only gravity does work: $mgh = \frac{1}{2}mv^2$, so $v = \sqrt{2gh} = \sqrt{2(9.80)(4.5)} = 9.4$ m/s. The mass cancels, and the slope angle does not matter for a frictionless path.

E2 (A). Energy with friction: $mgL(\sin \theta - \mu_k \cos \theta) = \frac{1}{2}mv^2$, so

$$v = \sqrt{2gL(\sin 30^\circ - 0.25 \cos 30^\circ)} = 3.73 \text{ m/s.}$$

Less than the frictionless $\sqrt{2gL \sin 30^\circ} = 4.95 \text{ m/s}$, as predicted.

E3 (D). $Q = f_k L = \mu_k mg \cos \theta L = (0.25)(3.0)(9.80)(0.866)(2.5) = 15.9 \text{ J}$. Check with energy: $mgL \sin \theta = 36.7 \text{ J} = K + Q = 20.8 + 15.9 \text{ J}$.

E4 (A). The spring energy becomes kinetic energy: $\frac{1}{2}kx^2 = \frac{1}{2}mv^2$, so $v = x\sqrt{k/m} = 0.10\sqrt{4000} = 6.32 \text{ m/s}$. Check units: $\sqrt{(\text{N/m})/\text{kg}} = \text{s}^{-1}$, times metres.

E5 (E). I is true: the work depends only on the end points, and these coincide. II is true: at a minimum $dU/dx = 0$ and the force points back toward it when the particle is displaced (a maximum would be unstable). III is true: the force points down the slope of U . Answer: I, II and III.

E6 (D). $F_x = -dU/dx = -(6.0x - 6.0x^2) = -(3.0 - 1.5) = -1.5 \text{ N}$. The negative sign means the force points toward decreasing x , down the slope of U (check the slope of U there is positive). The value $U(0.50) = 0.50 \text{ J}$ is an energy, not a force.

E7 (D). At the top, the minimum condition is $n = 0$, so $mg = mv^2/R$ and $v_{\min}^2 = gR$. Energy from release to the top: $mgh = mg(2R) + \frac{1}{2}mv_{\min}^2$, so $h = 2R + R/2 = 2.5R = 20.0 \text{ m}$. The height $2R = 16 \text{ m}$ would leave the car with no speed at the top, so it would leave the track.

E8 (B). The bob falls through $h = L(1 - \cos 45^\circ) = 0.351 \text{ m}$. Energy: $mgh = \frac{1}{2}mv^2$, so $v = \sqrt{2gh} = 2.62 \text{ m/s}$. The tension does no work (perpendicular to the path).

E9 (B). At the lowest point the net upward force provides the centripetal force: $T - mg = mv^2/L$, so $T = m(g + v^2/L) = 0.50(9.80 + 5.74) = 7.77 \text{ N}$. Greater than $mg = 4.90 \text{ N}$, as predicted. Here $v^2 = 6.89 \text{ m}^2/\text{s}^2$ from the previous item.

E10 (B). Initial mechanical energy $mgH = 196 \text{ J}$; final $\frac{1}{2}mv^2 = 144 \text{ J}$. The difference $196 - 144 = 52 \text{ J}$ went into thermal energy of the air and object. The no-resistance speed would be $\sqrt{2gH} = 14.0 \text{ m/s}$, larger than 12.0 , consistent with the loss.